

College Council & Club Management System

CSE-302: Assignment-2

Entity-Relationship (ER) Diagram Report

Group 4

Name	Roll Number
Ritesh Kuila	24075074
Vaibhav Aditya	24075092
Ghanendar	24075027
Shashwat Pratap Singh	24075080
Kamlesh Kumawat	24074021
Abhinav Anand	24075002

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1 Overview

This report presents the Entity-Relationship (ER) diagram for the College Council & Club Management System, based on the requirement specification and database schema described in the group's Assignment-1 report. The core entities represent the hierarchical structure of the college's student organizations (Council $\rightarrow$ Club $\rightarrow$ Vertical) along with the resources and activities they interact with (Student, Room, Event).

2 Entities and Attributes

- **Council:** CouncilID (primary key), Name, Description
- **Club:** ClubID (primary key), Name – belongs to one Council
- **Vertical:** VerticalID (primary key), Name – belongs to one Club
- **Event:** EventID (primary key), Name, EventDate, AcademicYear – organized by one Club
- **Student:** RollNumber (primary key), Name, GraduationYear, Branch, Email
- **Room:** RoomID (primary key)

3 Relationships

- **Contains:** Council – Club (1:N). A council contains many clubs; a club belongs to exactly one council.
- **Has:** Club – Vertical (1:N). A club has many verticals; a vertical belongs to exactly one club.
- **Organizes:** Club – Event (1:N). A club organizes many events; an event is organized by exactly one club.
- **Membership:** Student – Club (M:N), carrying the attributes PORTitle and AcademicYear. A student may hold memberships in multiple clubs, and a club has many student members.
- **Allocated:** Club – Room (M:N), carrying the attribute AcademicYear. A club may be allocated one or more rooms, and a room may be shared by more than one club.
- **Involved:** Student – Event (M:N), carrying the attributes RoleCategory, SpecificRole, and ResultDetails. A student may be involved in many events (as a participant, organizer, or volunteer), and an event involves many students.

4 Entity-Relationship Diagram

Figure 1 shows the ER diagram for the College Council & Club Management System.

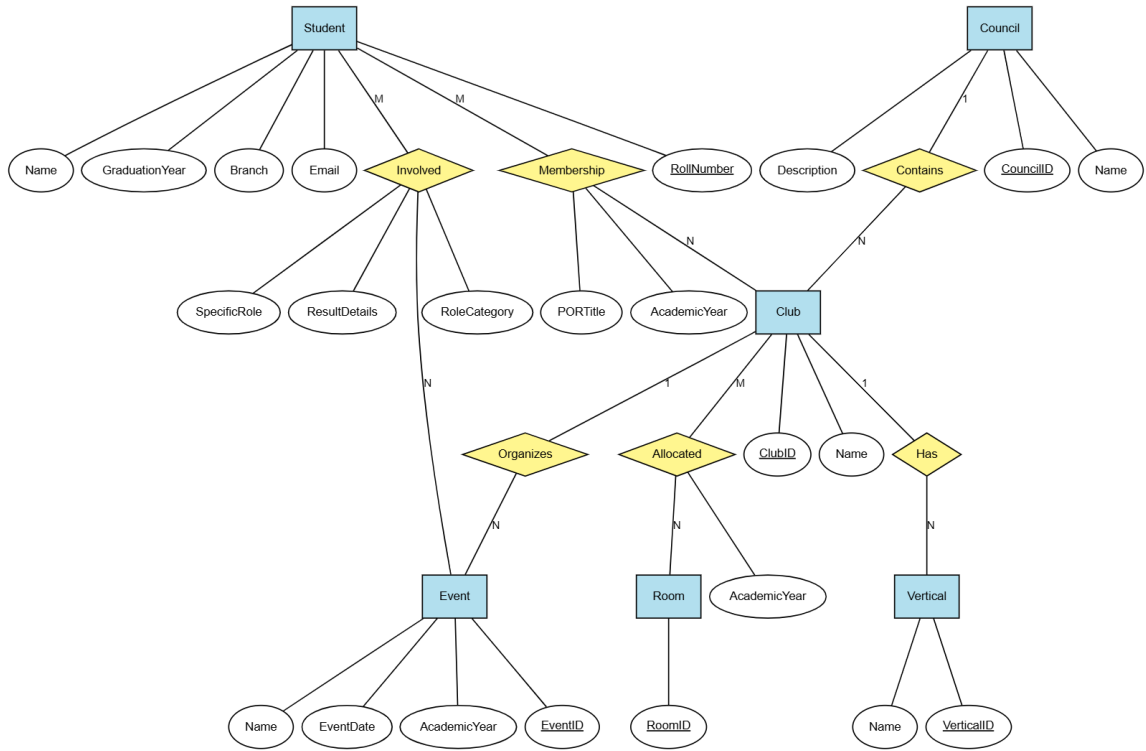


Figure 1: ER Diagram for the College Council & Club Management System

5 Conclusion

The ER diagram captures the hierarchical organization of college councils, clubs, and verticals, along with the many-to-many relationships between students and the clubs, events, and rooms they interact with. All many-to-many relationships that carry additional data (membership tenure and role, room allocation year, and event involvement details) are resolved into their own mapping tables at the relational schema stage, ensuring the database remains normalized and free of redundancy.